Introduction to Programming and Algorithms in Python

Syllabus • Tel Aviv University • Oct 20, 2026 – Jan 18, 2027

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Credits	4 semester credits
Language of instruction	English
Prerequisites	None. No prior programming experience is assumed.
Course website	`https://drorjac.github.io/python-course/`
Graduation requirements	Final exam, 3 homework submissions (HW3 is the final project, with a short presentation) and class participation.
Grading	Final exam 50%, homework 40% (HW1 10%, HW2 10%, HW3 final project 20%), class participation 10%.

Course Description

An introductory course in programming and algorithms in Python: basic programming, data structures, functions, scientific packages for data analysis and visualization, and an introduction to AI and machine learning. The course has 10 lecture units in 4 topics, taught over the semester, with a recitation and an enrichment session for each topic. Recitations include more practice material than is covered in class, for self-study at home. Programming environments: Google Colab, PyCharm and Visual Studio Code.

Syllabus

Topic 1: Intro and Python Basics (Lectures 1–3)

- **Lecture 1 – Introduction to Programming:** programs and algorithms, Python and its uses, AI and large language models (LLMs) in plain words, programming environments, print and comments.
- **Lecture 2 – Variables and Data Types:** variables and assignments, data types (int, float, string, boolean), arithmetic operations.
- **Lecture 3 – Input and Output:** input/output, type conversion, formatted output (f-strings), introduction to conditions (if-else).

Recitation 1: practice for Lectures 1–3 (partly in class, the rest for practice at home).

HW1: 'Introduction'. All of Topic 1: print and comments, variables, types and arithmetic, input and type conversion, f-strings, and basic conditions (`if` / `else`). Fixing bugs, completing missing code and writing short programs in a Google Colab notebook. *Given in early November, due ∼Nov 20.*

Topic 2: Decisions and Loops (Lectures 4–6)

- **Lecture 4 – Conditions:** if, elif, else, logical operators, binary logic (0/1, logic gates).
- **Lecture 5 – Lists and Sets:** data collections and built-in methods, string and list indexing and slicing.
- **Lecture 6 – Loops:** iteration using while and for, basic algorithms (sum, average, count, maximum, search).

Recitation 2: practice for Lectures 4–6 (partly in class, the rest for practice at home).

Topic 3: Functions and Dictionaries (Lectures 7–8)

- **Lecture 7 – Functions:** modular code, defining and calling functions, parameters and return values, local vs. global scope.

HW2: 'Functions and Loops'. Conditions, loops, lists, sets, string operations and basic functions to solve algorithmic problems. *Given in early December, due ∼Dec 20.*

- **Lecture 8 – Dictionaries and Debugging:** dictionaries and data mapping, nested lists and matrices, median and quantile, debugging, error handling with `try` / `except`, working with AI coding assistants.

Recitation 3: practice for Lectures 7–8 (partly in class, the rest for practice at home).

Topic 4: Data Science and AI (Lectures 9–10)

- **Lecture 9 – Objects, NumPy and Pandas:** a first look at object-oriented programming (OOP): objects and classes (`class`, `__init__`, `self`, methods), arrays and vectorized operations, image processing with NumPy, DataFrames, CSV datasets and basic statistics.
- **Lecture 10 – Plots and Machine Learning:** Matplotlib visualization, scikit-learn introduction (regression, classification, clustering), tools for the next step: an IDE, a debugger and AI coding assistants (demo), course summary.

Recitation 4: practice for Lectures 9–10 (partly in class, the rest for practice at home).

HW3 – Final Project: 'Data Analysis and Visualization'

The third homework is the final project. It brings the whole course together (with a focus on Lectures 8–10): loading a dataset, organizing it with dictionaries, tables and a small class of your own, basic statistical analysis with NumPy and Pandas, and visualization with Matplotlib. Each student presents the project briefly in the last week. *Given in January, due in the last week of the semester.*

Submission Schedule

Assignment	Covers	Weight	Due (approx.)
HW1: Introduction	Topic 1 (Lectures 1–3, incl. `if` / `else`)	10%	Nov 20
HW2: Functions and Loops	Topics 2–3 (Lectures 4–7)	10%	Dec 20
HW3: Final project + presentation	The whole course, focus on Lectures 8–10	20%	Jan (last week)

One homework a month: HW1 in November, HW2 in December, HW3 in January. Each homework is a Jupyter notebook (`.ipynb`): work in Google Colab, then download the notebook and submit it on Moodle. Each homework gets one overall grade. Exact dates and the full homework instructions are on Moodle.

Use of AI Tools

AI assistants (ChatGPT, Claude, Gemini, Copilot, ...) are part of programming today, and we use them in class. The rule is simple: **AI may help you learn, but the work you hand in must be yours.**

- **Allowed:** asking AI to explain an idea, an error message or a piece of code; asking for extra examples and practice questions; asking for a hint when you are stuck; checking your own solution.
- **Not allowed:** pasting the assignment into an AI and handing in its answer; handing in code you cannot explain line by line; using AI during the final exam.

- **Say how you used it:** at the top of every homework file, add a short comment, e.g. `# AI use: asked ChatGPT to explain a KeyError` or `# AI use: none`.
- **You may be asked to explain your code.** If you cannot explain code you submitted, it is treated as work that is not your own.
- **AI makes mistakes.** You are responsible for every line you submit: run it and test it.

This syllabus may be updated during the semester. Changes will be announced in class and on Moodle.